Comparativa Tiempo Real y Propuesta de Planificación

1. Propuesta de Planificación.

Realizar la planificación de las siguientes tareas en el simulador usando los métodos de Prioridades Estáticas y Dinámicas que sean aplicables.

Tareas	C	D	T
A	3	11	12
B	4	21	24
C	8	19	24
D	11	45	48

Realizamos las diversas planificación:

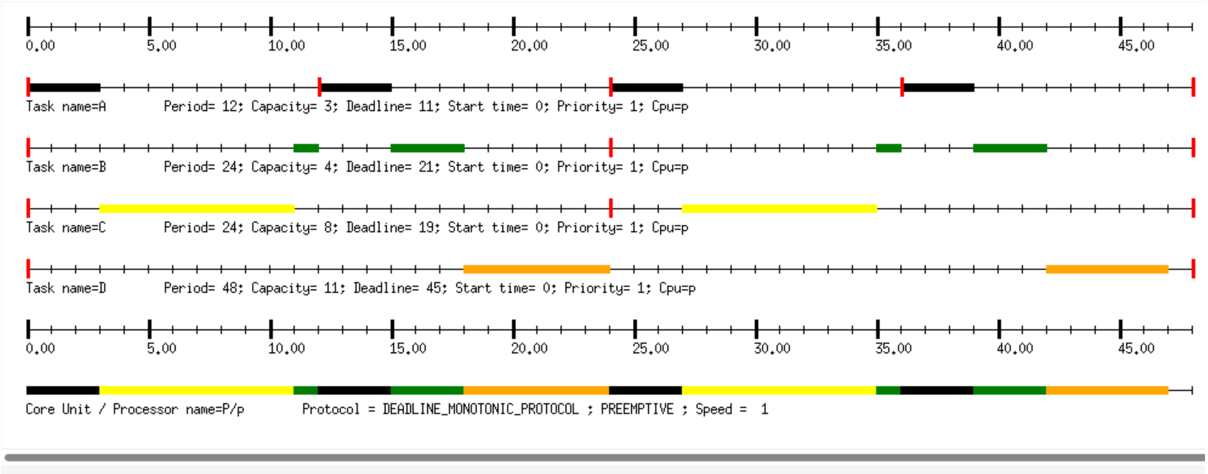
RM: vemos que no podemos pues:

Scheduling simulation, Processor p :
- Task must have period equal to deadline : cannot apply the selected scheduler with this task set.

- Task must have period equal to deadline : cannot apply the selected scheduler with this task set.

- Task response time computed from simulation :
  A => 0/worst , response time not computed since the task did not run all its capacity
  B => 0/worst , response time not computed since the task did not run all its capacity
  C => 0/worst , response time not computed since the task did not run all its capacity
  D => 0/worst , response time not computed since the task did not run all its capacity

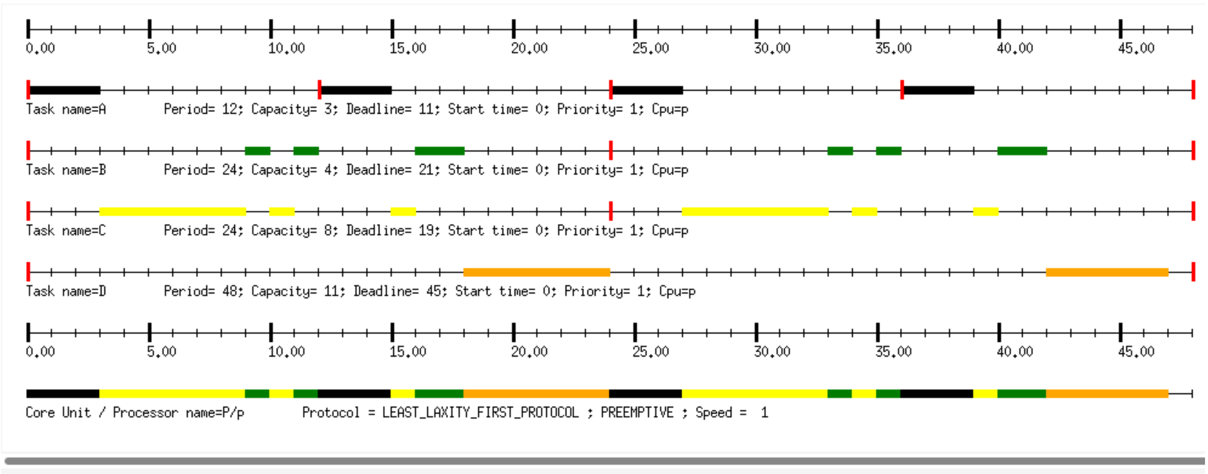
DM



Scheduling simulation, Processor p :
- Number of context switches : 11
- Number of preemptions : 3

- Task response time computed from simulation :
  A => 3/worst
  B => 18/worst
  C => 11/worst
  D => 47/worst , missed its deadline (absolute deadline = 45 ; completion time = 47)
- Some task deadlines will be missed : the task set is not schedulable.

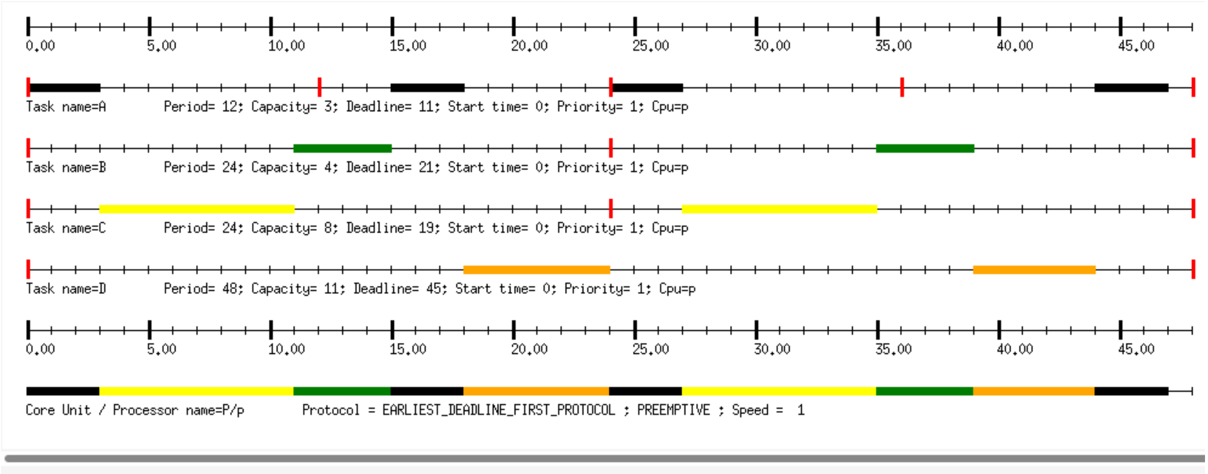
LLF



Scheduling simulation, Processor p :
- Number of context switches : 17
- Number of preemptions : 9

- Task response time computed from simulation :
  A => 3/worst
  B => 18/worst
  C => 16/worst
  D => 47/worst , missed its deadline (absolute deadline = 45 ; completion time = 47)
- Some task deadlines will be missed : the task set is not schedulable.

EDF



Scheduling simulation, Processor p :
- Number of context switches : 9
- Number of preemptions : 1

- Task response time computed from simulation :
  A => 11/worst
  B => 15/worst
  C => 11/worst
  D => 44/worst
- No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

a) ¿Qué método funciona mejor para cada tarea teniendo en cuenta los tiempos de respuesta en el mejor y peor caso? ¿Qué método funciona mejor para todas las tareas?

Task	Mejor caso	Peor caso
A	DM / LLF	EDF
B	EDF	DM / LLF
C	DM / EDF	LLF
D	EDF	DM / LLF

Él método que mejor funciona para todas las tareas es el EDF pues el que consigue planificarlas todas consiguiendo así que todas cumplan sus plazos de ejecución y además con menor número de cambios de contexto.

b) Para el método de Prioridades Estáticas empleado, calcular con el simulador si se podría planificar usando los test de Factor de Utilización y Tiempos de Respuesta. ¿Los resultados al simular han sido los esperados?
Con el RM directamente no se puede planificar porque no se cumple el requisito de que el periodo sea igual al deadline.

Scheduling feasibility, Processor p :
1) Feasibility test based on the processor utilization factor :

- The feasibility interval is 48.0, see Leung and Merill (1980) from [21].
- 1 units of time are unused in the feasibility interval.
- Number of cores hosted by this processor : 1.
- Processor utilization factor with deadline is 1.12870 (see [1], page 6).
- Processor utilization factor with period is 0.97917 (see [1], page 6).
- In the preemptive case, with DM, the task set is not schedulable because the processor utilization factor 1.12870 is more than 1.00000 (see [7]).

2) Feasibility test based on worst case response time for periodic tasks :

- Worst case task response time : (see [2], page 3, equation 4).
  A => 3
  C => 11
  B => 18
  D => 47, missed its deadline (deadline = 45)
- Some task deadlines will be missed : the task set is not schedulable.

Usando el DM Protocol observamos que tiene un factor de utilización de periodo 0.97917. Este valor al ser mayor que el valor de utilización mínima pero menor que uno no se puede garantizar ni que se puede planificar ni que se sobrecarga. Así pues según las ecuaciones de recurrencias vemos que no es planificable pues su tiempo de ejecución es mayor al de su Deadline.

2. Precedencia de Tareas

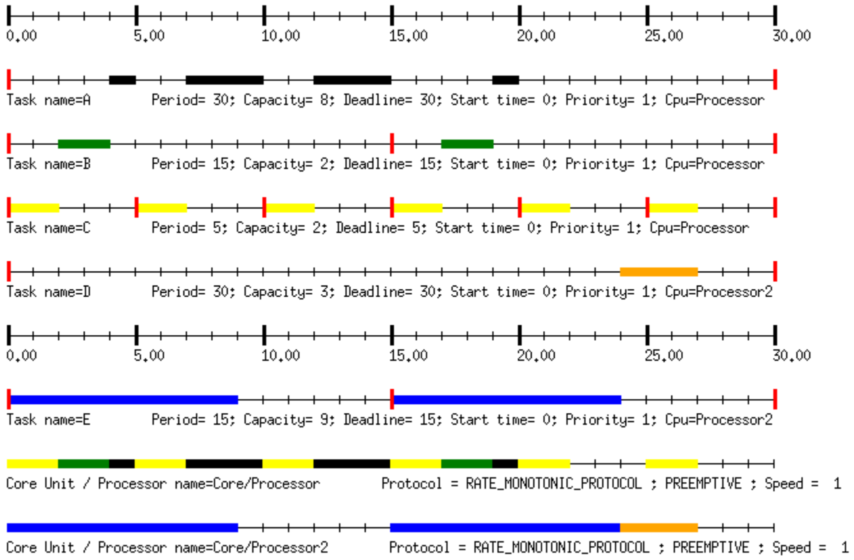
Dado el siguiente conjunto de tareas en un sistema multiprocesador

Tareas	C	T	Proc
A	8	30	1
B	2	15	1
C	2	5	1
D	3	30	2
E	9	15	2

a) Si la tarea A precede a la tarea D, realizar la planificación usando los siguientes métodos: RM, EDF y LLF (usando el mismo método de planificación para los dos procesadores). ¿Es planificable con todos los métodos? Si es así, ¿qué método elegirías para hacer la planificación?

Para la realización de de este ejercicio he utilizado un mismo core con dos procesadores:
Ejecuciones:

RM



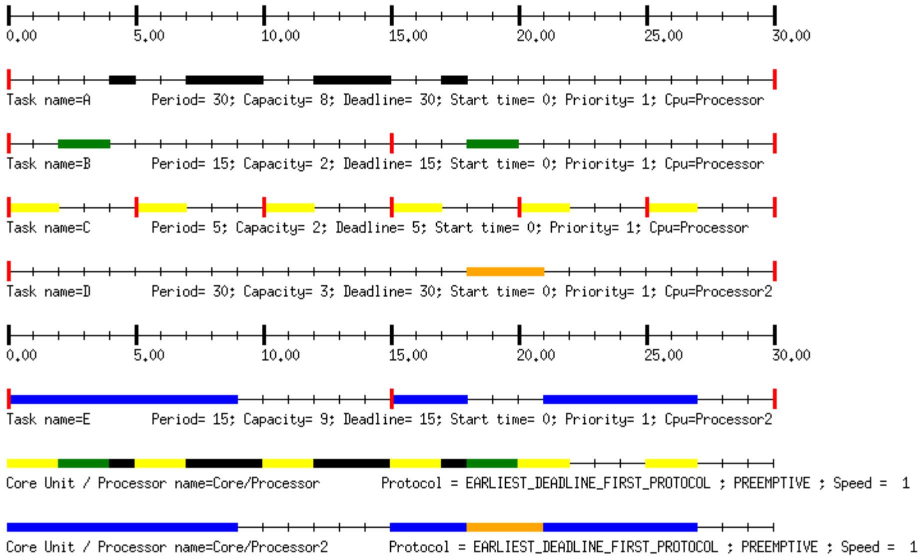
Scheduling simulation, Processor Processor :
Number of context switches : 10
Number of preemptions : 3

Task response time computed from simulation :
A => 20/worst
B => 4/worst
C => 2/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Scheduling simulation, Processor Processor2 :
Number of context switches : 1
Number of preemptions : 0

Task response time computed from simulation :
D => 27/worst
E => 9/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

EDF



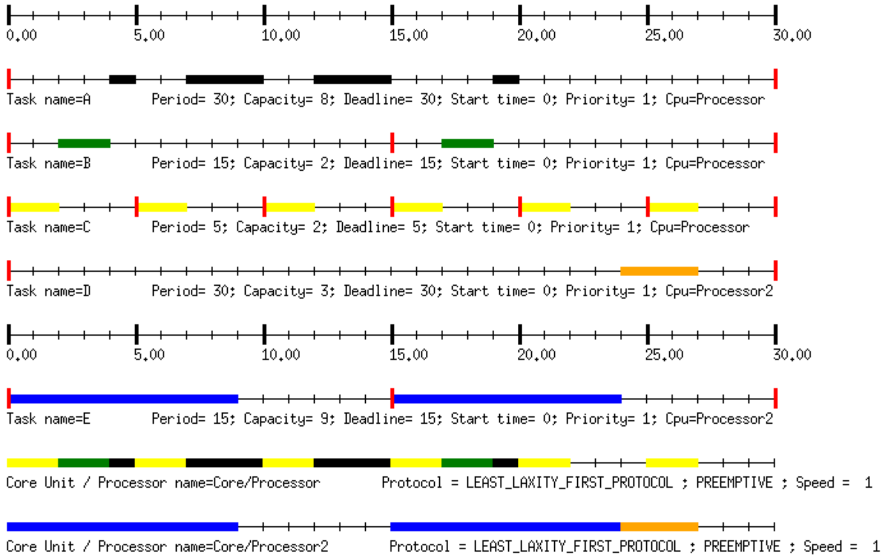
Scheduling simulation, Processor Processor :
Number of context switches : 10
Number of preemptions : 3

Task response time computed from simulation :
A => 18/worst
B => 5/worst
C => 2/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Scheduling simulation, Processor Processor2 :
Number of context switches : 2
Number of preemptions : 1

Task response time computed from simulation :
D => 21/worst
E => 12/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

LLF



Scheduling simulation, Processor Processor :
Number of context switches : 10
Number of preemptions : 3

Task response time computed from simulation :
A => 20/worst
B => 4/worst
C => 2/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Scheduling simulation, Processor Processor2 :
Number of context switches : 1
Number of preemptions : 0

Task response time computed from simulation :
D => 27/worst
E => 9/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Como podemos observar es planificable con todos los métodos. Veamos ahora cuál sería el mejor método a escoger viendo el worst task response de cada método y de cada task:

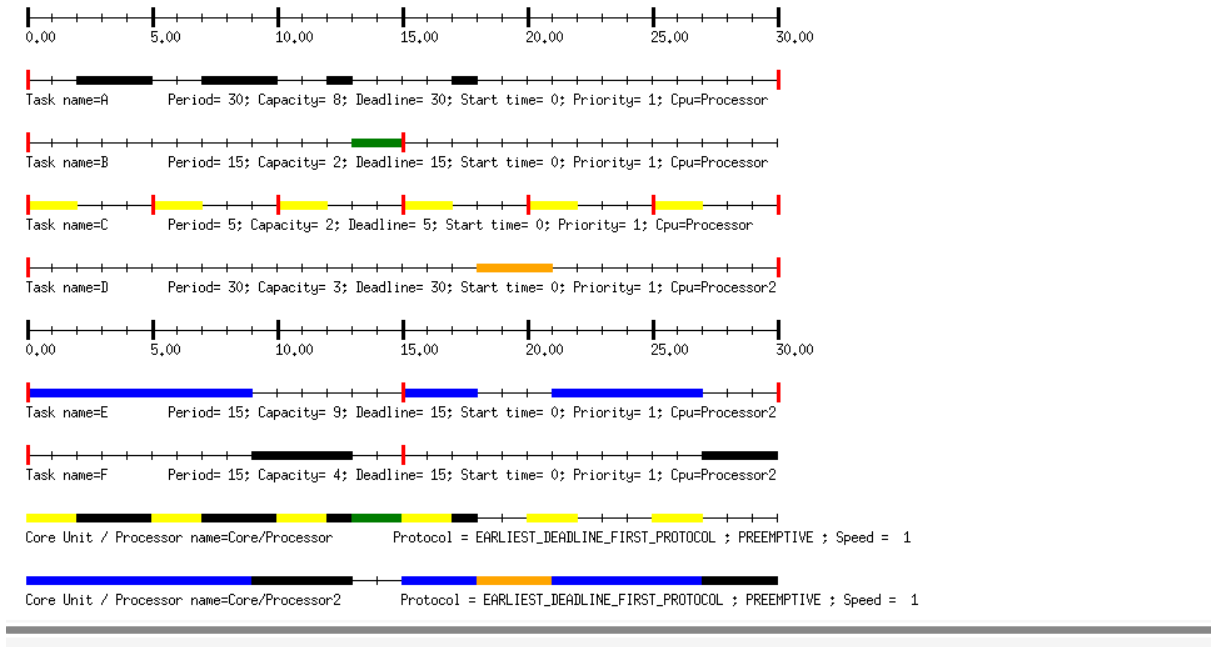
	A	**B**	**C**	**D**	**E**	**Total Task Response Time**
RM	20	4	2	27	9	62
EDF	18	5	2	21	12	58
LLF	20	4	2	27	9	62

Por tanto vemos que el mejor método es el EDF.

b) Si se añade una nueva tarea F al procesador 2, con las siguientes características: C = 4 y

T = 15, siendo esta nueva tarea predecesora de la tarea B, encontrar un método que permita que el sistema siga siendo planificable.

Probando con diversos métodos finalmente tras añadir la nueva tarea vemos que con EDF sigue siendo planificable:



Scheduling simulation, Processor Processor :
Number of context switches : 9
Number of preemptions : 3

Task response time computed from simulation :
A => 18/worst
B => 15/worst
C => 2/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Scheduling simulation, Processor Processor2 :
Number of context switches : 5
Number of preemptions : 1

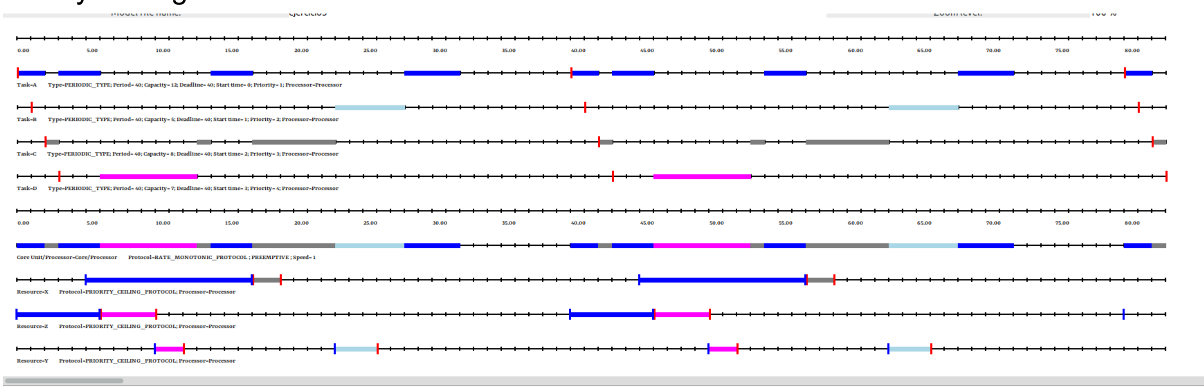
Task response time computed from simulation :
D => 21/worst
E => 12/worst
F => 13/worst
No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

3. Compartición de recursos

Dado el siguiente conjunto de tareas que comparten recursos realizar la planificación usando RM los diferentes protocolos de recursos y determinar en función de los tiempos de respuesta (para las tareas más prioritarias), cuál debería ser el protocolo elegido.
Nota: El 4 es la máxima prioridad.

Tareas	Ta	C	D	Prio	Recurso	Tbloqueo
A	0	12	40	1	X Z	5-8 1-5
B	1	5	40	2	Y	1-3
C	2	8	40	3	X	3-4
D	3	7	40	4	Y Z	5-6 1-4

Priority Ceiling Protocol

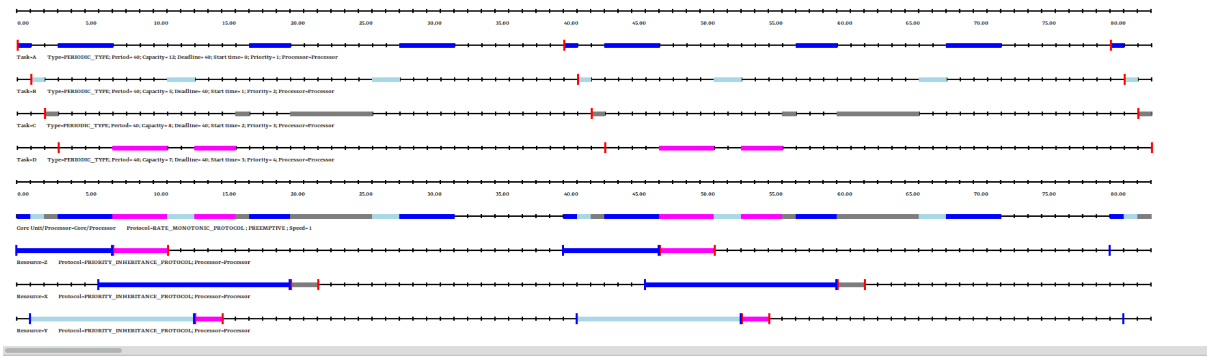


Scheduling simulation, Processor Processor :

- Number of context switches : 17
- Number of preemptions : 10

- Task response time computed from simulation :
  A => 32/worst
  B => 27/worst
  C => 21/worst
  D => 10/worst
- No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Priority Inheritance Protocol

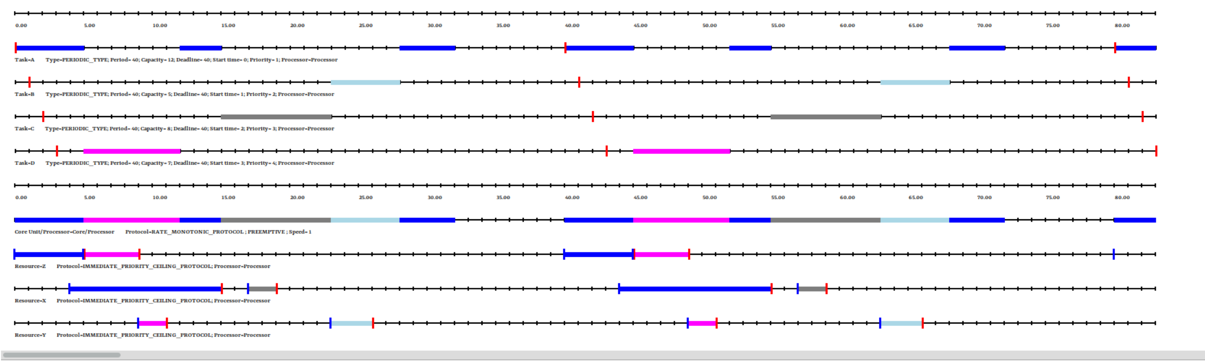


Scheduling simulation, Processor Processor :

- Number of context switches : 24
- Number of preemptions : 16

- Task response time computed from simulation :
  A => 32/worst
  B => 27/worst
  C => 24/worst
  D => 13/worst
- No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Inmediate Priority Ceiling Protocol



Scheduling simulation, Processor Processor :

- Number of context switches : 10
- Number of preemptions : 4

- Task response time computed from simulation :
  A => 32/worst
  B => 27/worst
  C => 21/worst
  D => 9/worst
- No deadline missed in the computed scheduling : the task set is schedulable if you computed the scheduling on the feasibility interval.

Para poder comparar los distintos resultados realizaré una tabla donde se visualicen:

	A	**B**	**C**	**D**	**Total Task Response Time**
Priority Ceiling Protocol	32	27	21	10	90
Priority Inheritance Protocol	32	27	24	13	96
Inmediate Priority Ceiling Protocol	32	27	21	9	89

Por tanto vemos que el mejor método es Inmediate Priority Ceiling Protocol (Techo Prioridad Inmediato).

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